MIL-C-45474B (MU)
30 December 1969
SUPERSEDING
MIL-C-45474A (MU)
13 March 1963

MILITARY SPECIFICATION

CHARGE, SPOTTING, FOR WARHEAD, M38

FOR

762MM ROCKET

LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of spotting charge designated as Charge, Spotting, for Warhead M38 for 762MM Rocket.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

FSC: 1340

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1234 - Pyrotechnics Sampling, Inspection and Testing
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

8833417 - Spotting Charge Loading Assembly.
8833418 - Metal Parts Assembly
8839586 - Carton, Packing, Ammunition, for Charge, Spotting, for M38 Warhead.
8839587 - Box, Packing, Ammunition, for Charge, Spotting, for M38 Warhead.

PUBLICATIONS

ARMY

ET-8833417 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

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CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 171-179

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify 49 CFR 171-179 (latest revision).)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Charge assemblies.-The charge assemblies shall comply with all requirements specified on Drawing (dwg.) 8833417 and with all requirements specified in applicable specifications.

3.3 Moisture content of charge.-The moisture content of the charge mixture at the loading station at the time of loading shall not exceed 0.10 percent.

3.4 Bonding.-The bonding surfaces shall be free of all foreign matter and the bonded joint shall be completely filled with adhesive.

3.5 Workmanship.-All parts must be free of dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent used. Surface coatings shall be continuous except for a few slight scratches not exposing base material. All marking shall be neat and sharply defined.

3.6 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.4)

3.7 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves

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the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing andspecification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-The contractor shall submit a sample of five (5) assemblies to a Government proving ground designated by the contracting officer for testing in accordance with the provisions of 4.2.2 (See 6.1). All parts and materials shall be the same as will be used in production and shall be obtained from the same source of supply. Prior to submission, the sample shall be inspected by the contractor to the degree necessary to assure that it conforms to the requirements of the contract.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity. Until the first article quantity is accepted, the contractor is not authorized by the Government to proceed with regular production unless otherwise directed by the Contracting Officer.

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4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Metal parts assemblies from lots bearing the same lot interfix number from one manufacturer.
b. Charge mixture from not more than one batch or homogeneous blend of batches.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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The absence of any inspection requirements in the specification shall not

4.3.2.1 Metal parts assembly (see dwg. 8833E18 covering a detail of dwg. 8833E17)

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical: None defined			
Major:	AQL 0.10 Percent		
101.	Internal coating damaged	Visual	01001
102.	Foreign matter on interior surface	Visual	01002
Minor: None defined			

4.3.2.2 Charge assembly, prior to closing (See dwg. 8833E17)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 Percent		
101.	Weight of charge	Balance	02001
102.	Foreign matter on bonding surface	Visual	02002
Minor: None defined			

4.3.2.3 Charge assembly (see dwg. 8833E17)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 Percent		
101.	Either cap insecure	Manual	03001
102.	Adhesive incomplete or not extending 360 degrees (see 3.4)	Visual	03002
103.	Disc color incorrect	Visual	03003
104.	Total length, max.	Gage	03007
Minor:	AQL 1.50 Percent		
201.	Marking missing, misleading or unidentifiable	Visual	03004
202.	Protective coating improper	Visual	03005
203.	Evidence of poor workmanship (see 3.5)	Visual	03006

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4.3.2.4 Carton, sealed (see dwg. 8839586)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Seal insecure	Visual/Manual	04001
102.	Contents loose	Manual	04002
Minor:	AQL 0.65 Percent		
201.	Marking missing, misleading or unidentifiable	Visual	04003

4.3.2.5 Barrier bag, sealed (see dwg. 8839586)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Air not exhausted	Visual	05001
102.	Seal inadequate	Manual	05002
Minor:	AQL 1.00 Percent		
201.	No room for subsequent seals	Visual	05003
202.	Marking misleading, missing, unidentifiable or improperly located	Visual	05004

4.3.2.6 Box, Packing (see dwg. 8839587)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 Percent		
101.	Contents exposed	Visual	06001
102.	FSC number missing or misleading	Visual	06002
103.	Board broken	Visual	06003
104.	Strapping missing, broken or loose	Visual	06004
105.	Contents loose	Manual	06005

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Minor:	AQL 1.00 Percent		
201.	Marking missing, misleading or unidentifiable	Visual	06006
202.	Paint missing	Visual	06007

4.3.3 Testing

4.3.3.1 Moisture content of charge mixture (see 3.3), Major Defect, Code No. 07001.-A sample of 20 gm. shall be selected from each batch or blend of batches for this test. If any sample fails to meet the requirement, the lot shall be rejected. The determination shall be performed as specified in 4.4.1 or 4.4.2.

4.3.3.2 Determination of aluminum content of charge batches (see dwg. 8833417) Major defect (Code No. 08001).-A sample of five grams (gm.) shall be selected from each batch or blend of batches for this test. If the sample fails to meet the requirement, the batch or blend shall be rejected. The determination shall be performed as specified in 4.4.3.

4.3.3.3 Vacuum chamber test (see dwg. 8839586) Major Defect, Code No. 09001.-Sampling and acceptance criteria shall be in accordance with inspection level S-4 of Standard MIL-STD-105. Test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 8833417 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Determination of moisture content.-Transfer to a tared glass-stoppered weighing bottle an accurately weighed portion of approximately 10 gm. of the charge mixture. Heat the bottle and contents in an oven at

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100 to 105 degrees centigrade (C.) for three hours, cool in a desiccator and weigh. Calculate the loss in weight as percent moisture. If a moisture determination is made on the ingredients prior to blending it is not necessary to perform this test provided the powder is to be loaded immediately. However, if the mixture has been shipped from place of manufacture to the loading facility or stored prior to loading, this determination shall be made.

4.4.2 Determination of moisture content (alternate method).-Method 101.2 of Standard MIL-STD-1234 may be used in lieu of the method required in 4.4.1.

4.4.3 Determination of aluminum content.-Transfer an accurately weighed portion of approximately two gms. of the sample to a tared medium porosity sintered glass crucible. Extract the sample with ten 25 milliliter portions of distilled water at a temperature of 0 to 5 degrees centigrade using a filtering flash and suction. Stir the contents of the crucible with each portion of water for one minute before applying suction. Rinse the undissolved residue first with alcohol, then with ether and aspirate until no further solvent odors are apparent. Place the crucible and contents in a drying oven at 100 to 105 degrees centigrade for 30 minutes, cool in a desiccator and weigh. Calculate the increase in weight of the crucible to percent aluminum on a moisture free basis as follows:

$$\text{Percent of aluminum} = \frac{100A}{W(1-M)}$$

Where:

A = increase in weight of filtering crucible.
W = weight of sample.
M = percent moisture expressed as a decimal fraction.

4.4.4 Vacuum chamber test.-The test procedure for this test shall be as prescribed in Specification MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-The assemblies shall be preserved and packaged in accordance with dwg. 8839586.

5.2 Packing

5.2.1 Level A.-The assemblies, preserved and packaged as specified in 5.1.1, shall be packed in accordance with dwg. 8839587.

5.3 Marking.-Marking shall be in accordance with dwgs. 8839586 and 8839587 and Code of Federal Regulations, Title 49 Parts 171-179.

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5.4 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication for procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When

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an equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (see 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the DL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally, within one month after receipt.

6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

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6.5 Exterior painting.-In the event a contract requires supply of metal parts and the loading thereof, exterior painting may be performed after loading if the painting procedure does not induce a static electrical charge in the assembly. The paint procedure shall be approved by the technical agency prior to use in production.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1340-A-

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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